



Why is there a difference between manually calculating a logistic regression 95% confidence interval, and using the confint() function in R?

Asked 13 years, 5 months ago Modified 5 years ago Viewed 43k times



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Dear everyone - I've noticed something strange that I can't explain, can you? In summary: the manual approach to calculating a confidence interval in a logistic regression model, and the R function `confint()` give different results.

I've been going through Hosmer & Lemeshow's *Applied logistic regression* (2nd edition). In the 3rd chapter there is an example of calculating the odds ratio and 95% confidence interval. Using R, I can easily reproduce the model:

```
Call:
glm(formula = dataset$CHD ~ as.factor(dataset$dich.age), family = "binomial")

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.734  -0.847  -0.847   0.709   1.549

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -0.8408     0.2551  -3.296  0.00098 ***
as.factor(dataset$dich.age)1  2.0935     0.5285   3.961  7.46e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 136.66  on 99  degrees of freedom
Residual deviance: 117.96  on 98  degrees of freedom
AIC: 121.96

Number of Fisher Scoring iterations: 4
```

However, when I calculate the confidence intervals of the parameters, I get a different interval to the one given in the text:

```
> exp(confint(model))
Waiting for profiling to be done...
              2.5 %      97.5 %
(Intercept)    0.2566283  0.7013384
as.factor(dataset$dich.age)1 3.0293727 24.7013080
```

Hosmer & Lemeshow suggest the following formula:

$$e^{[\hat{\beta}_1 \pm z_{1-\alpha/2} \times \text{SE}(\hat{\beta}_1)]}$$

and they calculate the confidence interval for `as.factor(dataset$dich.age)1` to be (2.9, 22.9).

This seems straightforward to do in R:

```
# upper CI for beta
exp(summary(model)$coefficients[2,1]+1.96*summary(model)$coefficients[2,2])
# lower CI for beta
exp(summary(model)$coefficients[2,1]-1.96*summary(model)$coefficients[2,2])
```

gives the same answer as the book.

However, any thoughts on why `confint()` seems to give different results? I've seen lots of examples of people using `confint()`.

r regression logistic confidence-interval profile-likelihood

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edited Apr 29, 2019 at 0:21



kjetil b halvorsen

79k 31 194 611

asked Dec 9, 2010 at 15:41



Andrew

6,248 5 25 22

- 2 Would you mind adding the exact literature reference for Hosmer & Lemeshow? I've been looking for the suggestion in their publications and books for quite some time, but haven't found it yet. – R大卫 Nov 7, 2014 at 8:02

3 Answers

Sorted by: Highest score (default)



After having fetched the data from the [accompanying website](#), here is how I would do it:

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```
chdage <- read.table("chdage.dat", header=F, col.names=c("id", "age", "chd"))
chdage$aged <- ifelse(chdage$age>=55, 1, 0)
mod.lm <- glm(chd ~ aged, data=chdage, family=binomial)
summary(mod.lm)
```



The 95% CIs based on profile likelihood are obtained with

```
require(MASS)
exp(confint(mod.lm))
```



This often is the default if the `MASS` package is automatically loaded. In this case, I get

```
                2.5 %      97.5 %
(Intercept) 0.2566283 0.7013384
aged        3.0293727 24.7013080
```

Now, if I wanted to compare with 95% Wald CIs (based on asymptotic normality) like the one you computed by hand, I would use `confint.default()` instead; this yields

```
                2.5 %      97.5 %
(Intercept) 0.2616579 0.7111663
aged        2.8795652 22.8614705
```

Wald CIs are good in most situations, although profile likelihood-based may be useful with complex sampling strategies. If you want to grasp the idea of how they work, here is a brief overview of the main principles: [Confidence intervals by the profile likelihood method, with applications in veterinary epidemiology](#). You can also take a look at Venables and Ripley's *MASS* book, §8.4, pp. 220-221.

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edited Jan 22, 2017 at 23:19



Askur

105 4

answered Dec 9, 2010 at 19:00



chl

53.9k 23 223 383

- ▲ The parameters of the binomial model are on the logit scale. I think you should be using `inv.logit` rather than `exp` to convert back in to the probability scale? – Greg Sexton Apr 8, 2021 at 9:59

- ▲ The inverse logit is defined as $\exp(x)/(1+\exp(x))$. I don't think this is what we want here. – chl Feb 14, 2022 at 21:14



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Following up: profile confidence intervals are more reliable (choosing the appropriate cutoff for the likelihood does involve an asymptotic (large sample) assumption, but this is a much weaker assumption than the quadratic-likelihood-surface assumption underlying the Wald confidence intervals). As far as I know, there is no argument for the Wald

statistics over the profile confidence intervals except that the Wald statistics are much quicker to compute and may be "good enough" in many circumstances (but sometimes way off: look up the Hauck-Donner effect).

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answered Dec 12, 2010 at 19:47

Ben Bolker

44k 3 119 167

- 2
- Thanks for this, and for suggesting I look up the Hauck-Donner effect. The effect doesn't get much treatment in textbooks but seems pretty important! – Andrew Nov 15, 2012 at 20:40

I believe if you look into the help file for confint() you will find that the confidence interval being constructed is a "profile" interval instead of a Wald confidence interval (your formula from HL).

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answered Dec 9, 2010 at 16:17

B_Miner
8,780 22 92 153

- 6
- Ahh. That answers the question. However, it leads on to the next one - which one is preferred? – Andrew Dec 9, 2010 at 17:37